98%

ABERDARE TECHNICAL COLLEGE

LEVEL 5,MODULE 3

CAT2,TERM 3

ELEMENT 2:TEST COMPUTER SOFTWARE FUNCTIONALITY

UNIT OF COMPETENCE: INSTALL COMPUTER SOFTWARE

UNIT CODE: 0619 451 06A

Total Marks: 36 **Time:** 1 Hour 30 Minutes

Name: Shirlene Karimi Reg. No: 962 Date: _____

Total Marks: 40 **Time:** 1 Hour 30 Minutes

QUESTION 1 (8 marks)

(a) Define the term software testing. **(2 marks)**

(b) Define each of the following terms: **(3 marks)**

(i) Error

(ii) Bug (defect)

(iii) Failure

(c) State any three HTTP methods used in REST and give the purpose of each. **(3 marks)**

QUESTION 2 (8 marks)

(a) Explain the difference between verification and validation, giving one example of each. **(4 marks)**

(b) Using the restaurant analogy (customer, waiter, kitchen), explain what an API is and how it works. **(4 marks)**

QUESTION 3 (8 marks)

(a) An age field on a registration form accepts values from 18 to 60 only. **(4 marks)**

(i) Use equivalence partitioning to identify the valid and invalid partitions. (2 marks)

(ii) Use boundary value analysis to list the six values you would test. (2 marks)

(b) A school website has the base URL https://school.com/api/. State the HTTP method and a suitable endpoint you would use in Postman to: **(4 marks)**

(i) Retrieve the list of students

(ii) Add a new student

(iii) Update a student's details

(iv) Delete a student record

QUESTION 4 (8 marks)

(a) Compare black box testing and white box testing under the following headings: knowledge of code, skills required, coverage, and what each type finds. **(4 marks)**

(b) A login form accepts passwords of 8 to 12 characters. Analyze how static testing and dynamic testing would each detect a missing check for the 12-character maximum, and explain why the two approaches complement each other. **(4 marks)**

QUESTION 5 (4 marks)

A bank is building a new system that must handle financial transactions with strong built-in security, while a sister company is building a public mobile app backend for simple data retrieval.

Evaluate which of REST or SOAP is better suited to each project. Justify your choices with at least two reasons per project. **(4 marks)**

i(a) Software testing is the process of checking and evaluating software to identify errors or defects and ensure that it meets the required specifications and works correctly.

b) i) Error
An error is a human mistake made during software development that produces an incorrect result.

ii) Bug (Defect)
A bug (defect) is a flaw or mistake in software that causes it to produce incorrect results or behave unexpectedly.

iii) Failure
A failure occurs when software does not perform its required function as expected during execution.

c) HTTP methods used in REST
GET - Used to retrieve or fetch data from a server.
POST - Used to send data to a server to create a new resource.
PUT - Used to update or replace an existing resource on a server.

2a) Verification - is the process of checking whether the software is being developed according to the specified requirements.
Example -
Reviewing a program code to check whether it follows the design specifications.

Validation - is the process of checking whether the software meets the user's needs and performs the intended task.
Example
Testing a registration system to confirm that users can register successfully.

b) API using the restaurant analogy
1. Customer - the customer represents the client application that requests a service.
2. Waiter - the waiter represents the API which carries requests and responses between the customer and kitchen.
3. Kitchen - the kitchen represents the server, which processes the request.

4 Process - The customer places an order with the waiter. The waiter takes it to the kitchen, and after the kitchen prepares the food, the waiter brings it to the customer. Similarly API sends requests to a server and returns the response to the client.

Q3 a) (i) Valid Partition
Age from 18 to 60

(ii) Invalid Partition
Age - below 18 and age above 60.

(iii) Boundary value analysis
Maximum valid age is 60 and the minimum valid age is 18.
The test values are:
1. 17 - Invalid (below minimum) boundary.
2. 18 - Valid (minimum boundary
3. 19 - Valid (it is above the minimum boundary.
4. 59 - Valid - it is just below the maximum boundary
5. 60 - Valid - it's the maximum boundary
6. 61 - Invalid - it is above the maximum

b) i) Retrieve - Retrieve the lists of students - HTTP method is GET and the endpoint is /students.

ii) Add a new student - HTTP method is POST and the endpoint is /students.

iii) Update a student's details - HTTP method is PUT and the endpoint is /students/:id)

iv) Delete a student record the HTTP method is DELETE and the endpoint is /students/:id)

Q4 a) Blackbox testing vs White box testing (Difference)
Blackbox testing has no knowledge of internal code required while white box testing has the knowledge of the internal code required.
Black box testing requires the skill of understanding requirements and test cases while white box testing requires the skill of programming and code analysis skills.
Black box testing tests functionality against requirements while white box testing tests statements, branches and code paths.

Black box testing finds missing functions, incorrect outputs, and interface errors while white box testing finds logic errors, coding errors, and unreachable code.

b) i) Static testing examines software code or documents without running the program.
A code review could reveal that the password validation does not check the maximum length of 12 characters.
Dynamic testing runs the program using test inputs.
Entering a password of 13 characters should be rejected. If it is accepted, the missing maximum-length is detected.
The reason they complement each other
ii) Static testing identifies potential defects before execution.
iii) Dynamic testing confirms whether the software behaves correctly during execution.
Together, they help detect more defects and improve software quality.

Qs 2) Bank financial transaction system - SOAP
Reasons:
1. SOAP supports security standards.
SOAP supports reliable messaging and formal transaction-related standards, making it suitable for suitable for systems requiring strong security and reliability
ii) Public mobile app backend for simple data retrieval - REST
Reasons
1. REST is lightweight and commonly uses JSON, reducing the amount of data exchanged.
REST uses standard HTTP methods such as GET to retrieve data making it simple and efficient for mobile applications.